

*Calculators are not allowed on this exam. Work all questions completely. Show all work as described in class.
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Point values for each problem are in the boxes in the margin.*

- 7 1. Set up an integral to find the area of the region bounded by $y = \sqrt{x}$ and $y = \frac{1}{3}x$.
2. Set up an integral to find the volume of the solid generated when the region bounded by $y = \sqrt{x}$ and $y = \frac{1}{3}x$ is
- 7 (a) Rotated about the x -axis using washers
- (b) Rotated about the x -axis using shells
- 7 (c) Rotated about the line $x = -4$
- 7 (d) Rotated about the line $y = 5$

- 14 3. Set up integrals to find the mass, the moment about the x -axis, and the moment about the y -axis of the region bounded by $y = \sqrt{x}$ and $y = \frac{1}{3}x$. Give the formula to find the coordinates of the center of mass. (Be sure to indicate which integral is which.)

- 2 4. **BONUS.** Set up an integral to find the moment about the line $x = -3$ of the region bounded by $y = \sqrt{x}$ and $y = \frac{1}{3}x$.

- 10 5. Graph the function $r = 3(1 - \cos \theta)$, $0 \leq \theta \leq \pi$ and set up an integral to find the area enclosed by the graph.

8 6. Consider the curve given by $f(x) = \sqrt{x}$, $0 \leq x \leq 3$. Set up an integral to find the length of this curve.

6 7. A spring is naturally 5 ft long and has spring constant $k = 70$ lb/ft. Set up an integral for the work done in stretching the spring from 6 ft long to 8 ft long.

12 8. A conical tank has a height of 8 feet and a radius of 7 feet. It is filled with a fluid having density 63 lb/ft³. Set up an integral to find the work done in pumping the fluid out of the tank if the tank is filled to a depth of 5 feet and the water must also be pumped out a 4 foot pipe sticking out the top of the tank.

9. If the amount of happiness while working a math test is constant, then the total joy produced during the test is given by

$$\text{Joy} = \left(\sqrt{\text{Happiness}} \right) * (\text{The Time The Happiness Lasts}).$$

In a certain 50 minute Calculus test, the Happiness was given by $H(t) = 1 - \frac{t}{50}$, where $0 \leq t \leq 50$.

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- (a) Describe in complete detail the fundamental idea of Chapter 6 and explain in complete detail how this fundamental idea can be applied to find the total Joy in the test.

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- (b) Set up an integral to find the total Joy in the test.